The due date for submitting this assignment has passed.
Due on 2026-04-01, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-01, 07:55 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

1) Consider the vector spaces V and functions $\langle .,. \rangle : V \times V \rightarrow \mathrm{R}$ defined as follows: ***1 point***

i) $V = \mathrm{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_1 - x_2y_1 - x_1y_2 + 2x_2y_2$.
ii) $V = M_{2\times 2}(\mathrm{R})$ and $\langle A, B \rangle = Tr(AB)$,
where $Tr(M)$ denotes the trace of a matrix M, i.e., the sum of the diagonal elements of M.
iii) $V = M_{2\times 1}(\mathrm{R})$ and $\langle A, B \rangle = Tr(AB^t)$,
where $Tr(X)$ denotes the trace of a matrix X, i.e., the sum of the diagonal elements of X.
Y^t denotes the transpose of matrix Y.
iv) $V = \mathrm{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_2 + x_2y_1$.

- [x] (i) is an inner product.
- [x] (ii) is an inner product.
- [ ] (iii) is an inner product.
- [ ] (iv) is an inner product.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(i) is an inner product.
(iii) is an inner product.

2) Consider two linear transformations T and S from $\mathrm{R}^2 \rightarrow \mathrm{R}^2$ defined as $T(x, y) = (2x + y, x + y)$ and $S(x, y) = (x + cy, x + 2y)$. Let A and B be matrix representations of linear transformations T and S with respect to the standard bases of R^2 respectively. ***1 point***
Consider the following statements: